

Regularization and Stability: Assignment 3

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Abstract—Supervised machine learning aims to train models that generalize well to unseen data. Classical learning theory explains generalization through empirical risk minimization, model complexity, and regularization. A complementary explanation is provided by algorithmic stability, which studies how sensitive a learning algorithm is to small changes in the training sample. This report investigates the relationship between regularization, stability, and generalization using ordinary least squares, ridge regression, and lasso regression. Ordinary least squares is used as an unregularized baseline, ridge regression is studied as the main regularized model, and lasso regression is included as an additional comparison based on L_1 regularization. The results show that ridge regression improves stability by reducing the sensitivity of the learned model to individual training observations, and that the best test performance is achieved when the model balances fit and stability.

I. INTRODUCTION

Supervised learning aims to train models that perform well on unseen observations, not only on the training sample. In the classical framework of empirical risk minimization, the learner chooses a prediction rule by minimizing the average loss over the training data. Vapnik's risk minimization framework [1] formalizes this idea by distinguishing the unknown expected risk from the empirical risk computed on the available sample. Since the data distribution is unknown, the empirical risk is used as a practical approximation of the true risk.

A central difficulty is that a model with very low training error may still generalize poorly. Classical learning theory explains this problem using model complexity, uniform convergence, and capacity measures such as VC dimension [1]. Regularization is a practical response to this issue. By adding a penalty term to the objective, regularization discourages overly large or overly flexible solutions and can reduce overfitting.

Algorithmic stability provides a complementary explanation. Instead of asking only how large the hypothesis class is, stability asks how sensitive the learning algorithm is to small changes in the training set. Bousquet and Elisseeff in [2] showed that if removing or replacing one training observation causes only a small change in the learned function or its loss, then the empirical and true risks can be close. This makes stability directly connected to generalization.

This report studies this connection empirically using linear regression models. Ordinary least squares (OLS) is used as the unregularized baseline. Ridge regression is the main model because its L_2 penalty has a direct theoretical connection to stability. Lasso regression is included as an additional extension to compare L_2 and L_1 regularization. The empirical analysis focuses on how the regularization parameter λ

affects training error, test error, coefficient norm, leave-one-out prediction change, and loss-based stability.

The remainder of this report is organized as follows. Section II presents the theoretical background on regularization, generalization, and algorithmic stability. Section III describes the datasets, models, and experimental methodology used to estimate stability and test error. Section IV presents the empirical results and analyzes the effect of regularization strength, model complexity, and dataset size. Finally, Section V summarizes the main findings and discusses the relationship between stability and generalization.

II. BACKGROUND AND RELATED WORK

A. Empirical Risk Minimization and Regularization

Let the training sample be

$$S = \{(x_1, y_1), \dots, (x_n, y_n)\}. \quad (1)$$

The goal is to learn a prediction function $f(x, w)$, where w is the parameter vector. The expected risk is

$$R(w) = \mathbb{E}_{(x,y)}[L(y, f(x, w))], \quad (2)$$

but this quantity cannot be directly computed because the data-generating distribution is unknown. The empirical risk is therefore used:

$$R_{\text{emp}}(w) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i, w)). \quad (3)$$

This is the standard empirical risk minimization principle [1]. However, minimizing only the empirical risk can lead to overfitting when the model is too sensitive to the particular training sample.

Regularization modifies the objective by adding a penalty on the size or complexity of the parameters. For L_2 -regularized learning, the objective is

$$\min_w \left[\frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i, w)) + \lambda \|w\|_2^2 \right]. \quad (4)$$

The parameter λ controls the strength of regularization. When λ is very small, the fitted model behaves similarly to the unregularized empirical-risk minimizer. When λ is large, the coefficients are strongly shrunk toward zero.

B. Ordinary Least Squares and Ridge Regression

For a design matrix $X \in \mathbb{R}^{n \times d}$ and response vector $y \in \mathbb{R}^n$, the linear regression model is

$$y = Xw + \varepsilon. \quad (5)$$

OLS estimates w by minimizing the residual sum of squares:

$$\hat{w}_{\text{OLS}} = \arg \min_w \|y - Xw\|_2^2. \quad (6)$$

When $X^T X$ is invertible, the closed-form solution is

$$\hat{w}_{\text{OLS}} = (X^T X)^{-1} X^T y. \quad (7)$$

OLS is optimal for minimizing the training residual sum of squares, but it can be unstable when the predictors are correlated or when $X^T X$ is ill-conditioned. Hoerl and Kennard in [3] showed that when the prediction vectors are not close to orthogonal, least-squares estimates can become highly sensitive, and adding a positive quantity to the diagonal of $X^T X$ can reduce this instability.

Ridge regression modifies the least-squares objective by adding an L_2 penalty:

$$\hat{w}_{\text{ridge}} = \arg \min_w \left\{ \frac{1}{n} \|y - Xw\|_2^2 + \lambda \|w\|_2^2 \right\}. \quad (8)$$

The corresponding closed-form solution is

$$\hat{w}_{\text{ridge}} = (X^T X + n\lambda I)^{-1} X^T y. \quad (9)$$

Adding $n\lambda I$ improves the conditioning of the estimation problem and shrinks the coefficient vector. This is why Ridge can reduce variance and improve stability, even though it introduces bias.

C. Lasso Regression

Lasso regression uses an L_1 penalty instead of an L_2 penalty:

$$\hat{w}_{\text{lasso}} = \arg \min_w \left\{ \frac{1}{2n} \|y - Xw\|_2^2 + \lambda \|w\|_1 \right\}, \quad (10)$$

where

$$\|w\|_1 = \sum_{j=1}^d |w_j|. \quad (11)$$

The L_1 penalty can set some coefficients exactly equal to zero. Therefore, Lasso performs regularization and implicit feature selection. Lasso is useful as an empirical comparison because it shows how a different regularization geometry affects coefficient shrinkage, test error, and leave-one-out sensitivity.

D. Algorithmic Stability

Let A_S be the model trained on the full training sample S , and let $A_{S \setminus i}$ be the model trained after removing the i -th observation. Uniform stability with parameter β requires

$$\sup_z |\ell(A_S, z) - \ell(A_{S \setminus i}, z)| \leq \beta. \quad (12)$$

A smaller β means that removing one training observation has a smaller effect on the loss. Therefore, a stable algorithm is less dependent on any single observation and is expected to have a smaller generalization gap.

The stability view is different from the classical capacity view. Classical learning theory focuses on the size or

complexity of the hypothesis class, while algorithmic stability focuses on how the algorithm reacts to perturbations of the training sample [2]. This distinction is important because two algorithms may search the same hypothesis class but have different stability properties.

E. Stability Bound for Ridge Regression

For regularized least-squares regression, the stability bound of the form

$$\beta \leq \frac{2\kappa^2 B^2}{\lambda n}. \quad (13)$$

In this expression, κ^2 is an upper bound on the squared feature norm or kernel value, B is a bound related to the response values, λ is the regularization strength, and n is the training sample size [2]. For a linear model with finite data, κ can be interpreted empirically as a bound on $\|x_i\|_2$, while B can be interpreted as a bound on $|y_i|$ within the observed sample.

This bound predicts two qualitative patterns:

$$\beta \propto \frac{1}{\lambda} \quad \text{and} \quad \beta \propto \frac{1}{n}. \quad (14)$$

Thus, increasing λ or increasing n should reduce instability. However, the bound is an upper bound, not an exact equality. In addition, the synthetic data are generated from a Gaussian distribution, which is not globally bounded. Therefore, the empirical analysis should be interpreted as a finite-sample check of the qualitative $1/(\lambda n)$ prediction rather than as a strict verification of all theorem assumptions.

III. DATASETS

A. Diabetes Dataset

The Diabetes dataset contains 442 observations and 10 numerical input features which are explained in Table I. This dataset is suitable for linear regression because the target is continuous and the predictors are numerical [6].

The dataset is also useful for studying regularization because some predictors are correlated. The correlation heatmap in Fig. 1 shows that `bmi`, `lbg`, `bp`, and `tch` have positive associations with the target, while `hdl` has a negative association. The heatmap also shows correlations among predictors, such as the correlation between `tc` and `ldl`. This is relevant because Ridge regression was designed to reduce instability in nonorthogonal predictor settings [3].

All input variables in the scikit-learn version of the Diabetes dataset are numerical and already scaled. This makes the experiment easier to interpret because changes in coefficient norm are not driven by arbitrary feature units. The dataset is moderate in size, so leave-one-out retraining is computationally feasible.

B. Synthetic Dataset

The synthetic dataset was generated to provide a controlled, strictly linear regression setting. The dataset consists of $n = 250$ observations and $d = 20$ numerical features. The feature matrix was sampled from a standard multivariate Gaussian distribution:

$$X \sim \mathcal{N}(0, I_{d \times d}). \quad (15)$$

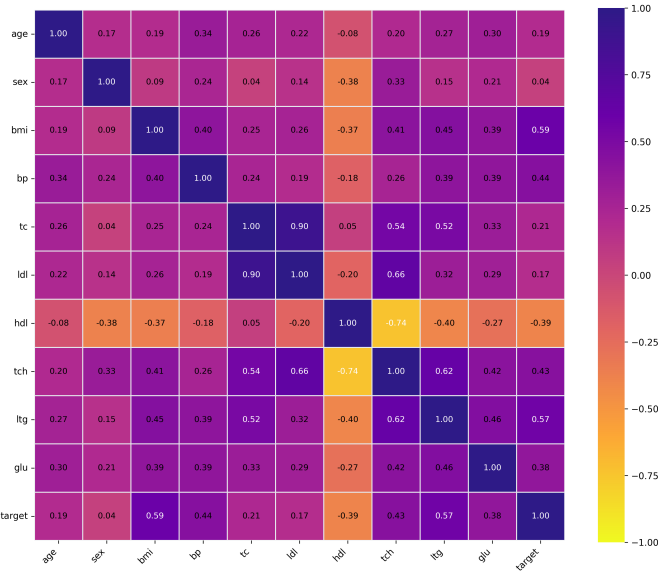


Fig. 1: Correlation heatmap of the Diabetes dataset features and the disease-progression target.

TABLE I: Dataset metadata table for the Diabetes dataset.

Variable	Role	Type	Description
age	Feature	Numeric	Age in years
sex	Feature	Numeric	Sex of the patient
bmi	Feature	Numeric	Body mass index
bp	Feature	Numeric	Average blood pressure
tc	Feature	Numeric	Total serum cholesterol
ldl	Feature	Numeric	Low-density lipoproteins
hdl	Feature	Numeric	High-density lipoproteins
tch	Feature	Numeric	Total cholesterol / HDL ratio
ltg	Feature	Numeric	Log serum triglycerides level
glu	Feature	Numeric	Blood sugar level
target	Target	Continuous	Disease progression after one year

The target variable was generated according to the true linear relationship:

$$y_i = x_i^T w^* + \epsilon_i. \quad (16)$$

To ensure exact reproducibility, the underlying data-generating parameters were explicitly defined using a fixed random seed. The ground-truth weight vector $w^* \in \mathbb{R}^{20}$ was sampled from a standard normal distribution, such that

$$w_j^* \sim \mathcal{N}(0, 1), \quad j \in \{1, \dots, 20\}. \quad (17)$$

The noise term ϵ_i was drawn from a normal distribution with variance $\sigma^2 = 25.0$, defined as

$$\epsilon_i \sim \mathcal{N}(0, 25). \quad (18)$$

This setting is particularly useful because the true data-generating process is completely known and linear. Therefore, OLS and Ridge regression are correctly specified models. This guarantees that the empirical effects of the regularization parameter λ can be studied in isolation, without model misspecification serving as a confounding variable.

Furthermore, this controlled synthetic framework allows for the systematic variation of the training sample size n , providing a direct empirical check of the theoretical prediction that algorithmic stability should improve as n increases.

IV. METHODOLOGY

This section describes the experimental procedure used to study the relationship between regularization strength, algorithmic stability, and generalization performance. The main comparison is between OLS, used as the unregularized baseline, and Ridge regression, evaluated across different values of the regularization parameter λ .

A. Model Fitting

In this experiment each dataset, 80% of the observations were used for training and the remaining 20% were held out for testing. The models were fitted on the training set and evaluated on the fixed test set. OLS was used as the baseline model because it does not include any regularization penalty. Ridge regression was then trained for several values of λ , ranging from very small values to large values.

Very small values of λ were included to show that Ridge behaves similarly to OLS when the regularization penalty is almost zero. Larger values of λ were included to study how stronger regularization affects coefficient shrinkage, stability, and test performance.

For each value of λ , the same training and test split was used. This makes the results comparable across different regularization strengths. The fitted models were evaluated using training mean squared error, test mean squared error, prediction-based stability, loss-based stability, and coefficient norm.

Lasso regression was also included as an additional regularized baseline. While Ridge uses an L_2 penalty and shrinks coefficients smoothly, Lasso uses an L_1 penalty and can set some coefficients exactly equal to zero. Therefore, Lasso was mainly used to compare how different regularization penalties affect coefficient shrinkage, model complexity, and test error. The main stability analysis, however, focuses on Ridge regression because the theoretical stability bound discussed in this report is directly connected to L_2 -regularized learning.

B. Stability Estimation

Empirical stability was estimated using a leave-one-out procedure. For a fixed value of λ , the model was first trained on the full training set. Then, each training observation was removed one at a time, and the model was retrained on the reduced training set.

After each retraining step, predictions were computed on the same fixed test set. The prediction-based stability estimate was computed as

$$\widehat{\text{stab}}(\lambda) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|T|} \sum_{x_t \in T} |f(x_t; S) - f(x_t; S \setminus i)|, \quad (19)$$

where S is the training set, $S \setminus i$ is the training set after removing the i -th observation, and T is the fixed test set. This measure represents the average change in model predictions caused by removing one training point.

A loss-based stability measure was also computed. Instead of comparing only the predictions, this measure compares the squared loss of the model trained on the full training set with the squared loss of the model trained after removing one observation. This is closer to the theoretical definition of algorithmic stability.

C. Evaluation Metrics

The mean squared error was used to measure prediction error:

$$\text{MSE}(D) = \frac{1}{|D|} \sum_{(x_i, y_i) \in D} (y_i - f(x_i; S))^2, \quad (20)$$

where D denotes the dataset on which the model is evaluated. When $D = S$, the metric gives the training MSE. When $D = T$, it gives the test MSE. Here, S is the training set, T is the fixed test set, and $f(x_i; S)$ is the prediction made by the model trained on S .

Prediction-based stability was computed using the leave-one-out procedure described in the previous subsection. It measures the average absolute change in predictions after removing one training observation.

The squared loss is defined as

$$\ell(f(x), y) = (y - f(x))^2. \quad (21)$$

Loss-based stability was measured by comparing the squared loss of the full model with the squared loss of the leave-one-out model:

$$\widehat{\text{Stab}}_{\text{loss}}(\lambda) = \frac{1}{n} \sum_{i=1}^n \frac{1}{|T|} \sum_{(x_t, y_t) \in T} \left| \ell(f(x_t; S), y_t) - \ell(f(x_t; S \setminus i), y_t) \right|, \quad (22)$$

where $n = |S|$.

Finally, model complexity was measured using the L_2 norm of the coefficient vector:

$$\|w\|_2 = \sqrt{\sum_{j=1}^d w_j^2}. \quad (23)$$

Together, these metrics make it possible to analyze the trade-off between fit, stability, and model complexity. An optimal model should have low test error and low sensitivity to small changes in the training data. However, very strong regularization may reduce the coefficient norm too much, causing the model to underfit. Therefore, the goal is not simply to maximize regularization, but to identify a useful balance between stability and predictive accuracy.

V. EVALUATION AND ANALYSIS

A. Effect of Regularization on Stability

The results in Fig. 2a and Fig. 2c show that Ridge regularization improves empirical stability as λ increases from very small to moderate values. For very small λ , Ridge behaves almost like OLS because the penalty term has almost no effect on the fitted coefficients. Therefore, the prediction-change values are close to the OLS baseline.

As λ increases, the average prediction change decreases. This means that removing one training observation has a smaller effect on the model predictions, so the model becomes significantly more stable. This pattern is especially clear in the synthetic regression dataset, where prediction change drops sharply after the regularization becomes strong enough. The Diabetes dataset shows the same general behavior, although the improvement is less smooth.

However, the improvement is not unlimited. In the Diabetes dataset, the lowest prediction change is obtained around a moderate value of λ , approximately $\lambda = 5$. After this point, the prediction-change value increases again for larger λ , although it remains below the OLS baseline. The loss-based stability curves in Fig. 2b and Fig. 2d show a similar non-monotonic behavior; loss change decreases when regularization is introduced, but very large values of λ no longer improve the result and can increase the stability metric again.

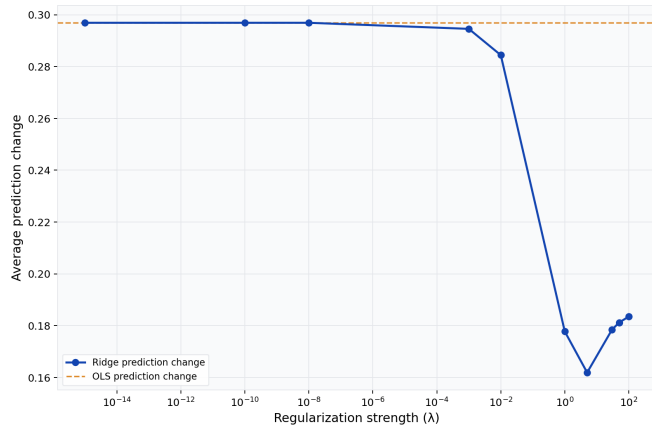
The Lasso curves shows a similar regularization effect; for very small values of λ , Lasso behaves close to OLS, while larger values of λ reduce the prediction-change and loss-change measures. This suggests that Lasso also improves empirical stability by restricting the fitted model. However, the main focus of this report remains Ridge regression, because the theoretical stability discussion is centered on the L_2 penalty.

This shows that stronger regularization improves empirical stability mainly up to a useful range. When λ becomes too large, the model is forced to be too simple because the coefficients are heavily shrunk toward zero. This effect is visible later in the model-complexity plots. In this region, the model can have lower sensitivity to individual observations, but it also loses predictive flexibility and begins to underfit the data.

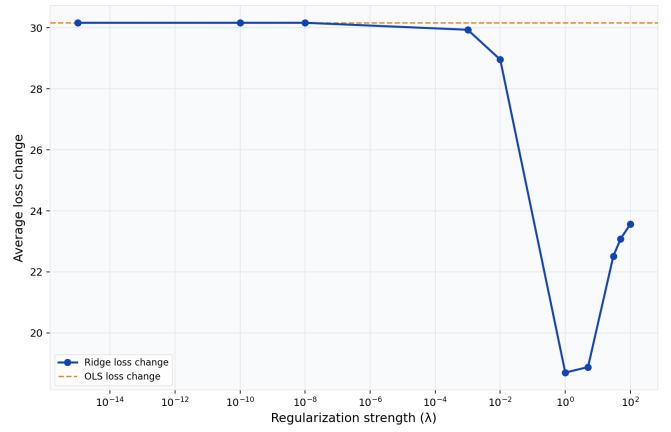
Overall, the figures support the theoretical expectation that Ridge regularization reduces sensitivity to individual observations relative to OLS. They also show the practical bias-variance trade-off: too little regularization gives a more sensitive model, while too much regularization causes underfitting and can worsen the stability-accuracy balance.

B. Test Error and the Bias-Variance Trade-Off

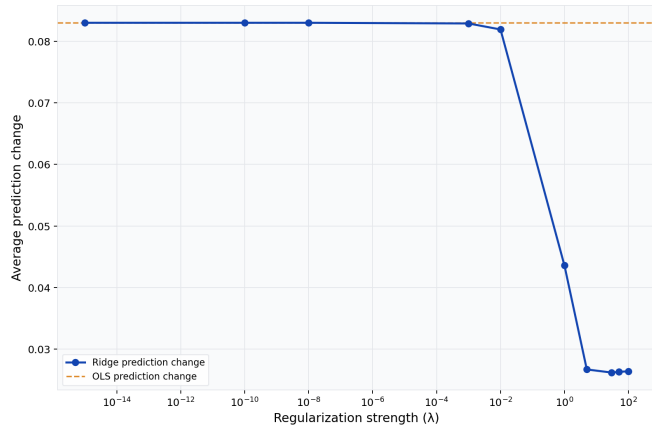
The train/test error curves in Fig. 3a and Fig. 3b show that stronger regularization does not necessarily improve predictive performance. For the Diabetes dataset, OLS and Ridge with very small λ values achieve the lowest or nearly lowest test MSE. The best Ridge result is obtained at a very small value of λ , which suggests that the unregularized model does not suffer from severe overfitting on this dataset.



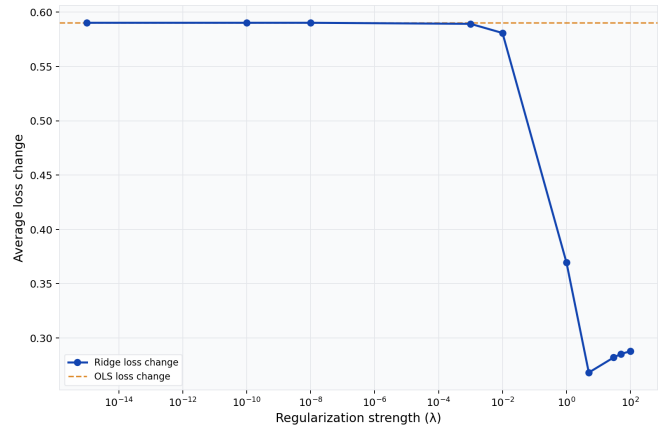
(a) Prediction stability for the Diabetes dataset.



(b) Loss stability for the Diabetes dataset.



(c) Prediction stability for the synthetic regression dataset.



(d) Loss stability for the synthetic regression dataset.

Fig. 2: Effect of regularization on prediction-based and loss-based stability for Ridge and Lasso on the Diabetes and synthetic regression datasets.

For the Diabetes dataset, Lasso achieves a slightly lower test MSE than Ridge for some regularization values in this run. However, the main pattern remains the same; very large regularization values increase the error because the fitted model becomes too restricted.

For the synthetic regression dataset, the same behavior is visible more clearly. Small values of λ perform similarly to OLS, while large values of λ increase both training and test error. This indicates underfitting. When the penalty is too strong, the coefficients are pushed too close to zero, and the model can no longer capture the signal in the data.

Therefore, the figures show the cost of excessive regularization. Moderate regularization can improve stability, but very strong regularization increases bias and reduces predictive accuracy.

C. Relationship Between Stability and Test Error

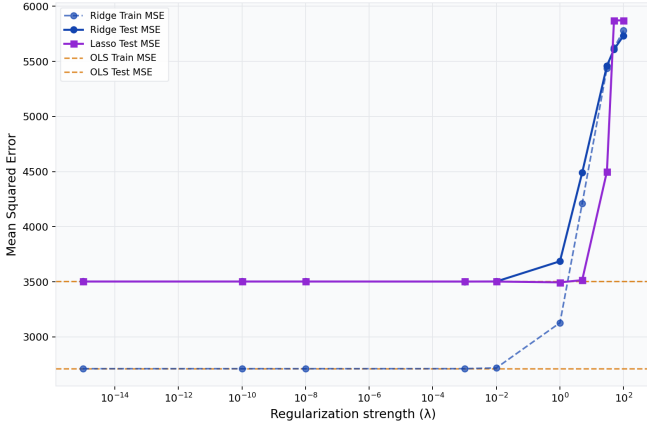
The plots in Fig. 4a and Fig. 4b summarize the trade-off between stability and predictive accuracy. Points closer to the lower-left corner are preferred because they have both lower prediction change and lower test MSE.

For the Diabetes dataset, the most stable Ridge model is not the model with the lowest test error. The best Ridge test MSE occurs at a very small value of λ , while the most stable Ridge model occurs around $\lambda = 5$. This shows that the most accurate model and the most stable model are not necessarily the same. Increasing λ improves stability, but after a point it increases test error.

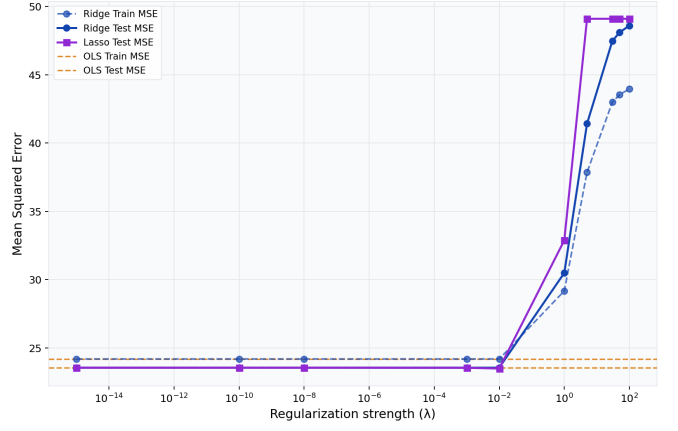
The same trade-off appears in the synthetic dataset. Larger λ values reduce prediction change up to the stable region, but they also move the model upward in test MSE once the regularization becomes too strong.

The Lasso points in Fig. 4a and Fig. 4b provide an additional comparison. Lasso also becomes more stable as regularization increases, but for large values of λ , it can suffer from the same underfitting problem. This is visible when the model moves toward lower prediction change but higher test MSE. Therefore, both Ridge and Lasso show the same general trade-off between stability and predictive accuracy.

These figures show that stability alone is not enough for good generalization. A useful model must be stable, but it must also remain flexible enough to fit the underlying relationship



(a) Training and test error curves for the Diabetes dataset.



(b) Training and test error curves for the synthetic regression dataset.

Fig. 3: Training and test error across different values of λ .

in the data.

D. Model Complexity

The model-complexity plots in Fig. 5a and Fig. 5b show that regularization reduces the size of the fitted coefficients. For Ridge regression, the L_2 norm of the coefficient vector decreases as λ increases. This confirms that the regularization penalty is shrinking the model coefficients.

The Lasso curves show a similar but stronger shrinkage effect. Because Lasso uses an L_1 penalty, it can produce sparse solutions by setting some coefficients exactly to zero. Although Fig. 5a and Fig. 5b show the coefficient norm rather than the number of nonzero coefficients, the sharp drop in the Lasso norm is consistent with this stronger shrinkage behavior.

These figures also help explain the error curves. When the coefficient norm becomes very small, the model becomes too simple and loses predictive power. This is visible by comparing Fig. 5a with Fig. 3a, and Fig. 5b with Fig. 3b. In both datasets, large λ values strongly reduce coefficient size, but this comes with a clear increase in test error.

E. Effect of Dataset Size

The synthetic dataset was also used to study the effect of training sample size on stability. For a fixed value of $\lambda = 1$, theory predicts that stability should improve as the number of training observations increases:

$$\beta \propto \frac{1}{n}. \quad (24)$$

The dataset-size experiment supports this expectation. In Fig. 6a, the average prediction change decreases steadily as the training sample size increases. This means that when more observations are available, removing one individual point has a smaller effect on the fitted Ridge model. Therefore, the model becomes more stable.

The test-error curve in Fig. 6b also shows an overall improvement as the training set becomes larger. Test MSE is highest for the smallest training size and decreases for larger

training sizes. Overall, Fig. 6a strongly supports the theoretical relationship between sample size and stability, while Fig. 6b shows that increasing the amount of training data also tends to improve generalization in this experiment.

VI. DISCUSSION

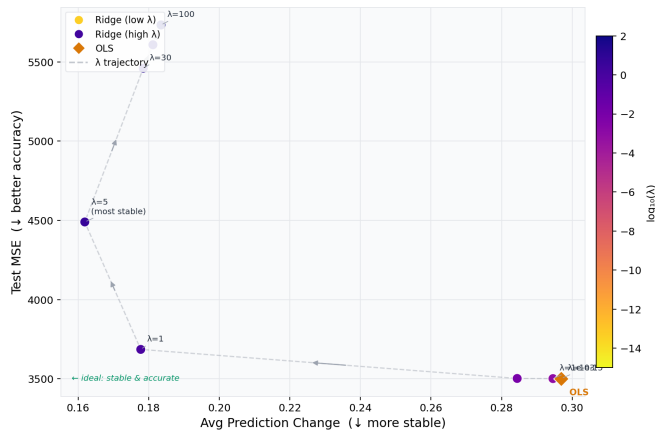
Ridge regression improves empirical stability as λ increases from very small to moderate values, but this improvement does not always correspond to better test performance.

This result shows that stability is useful but should not be treated as the only objective. A model can be stable because it is robust to small changes in the training set, but it can also be stable because it is too simple. In the large- λ region, Ridge coefficients are strongly shrunk toward zero, the coefficient norm becomes very small, and the model underfits. This explains why test error increases even though prediction change remains relatively low.

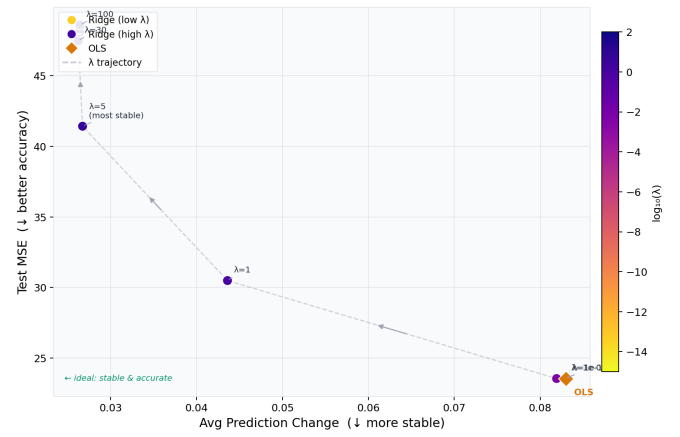
The Lasso results strengthen this interpretation. Lasso achieves the best test MSE on the Diabetes dataset at $\lambda = 1$, where it keeps 7 nonzero coefficients. This suggests that some feature selection can be useful. However, at larger λ values, Lasso removes all features and becomes an intercept-only model. This produces low sensitivity but poor prediction. Therefore, the most stable Lasso model is not the best model in a practical sense.

The synthetic dataset shows the same pattern in a more controlled setting. OLS and very weakly regularized Ridge perform best in terms of test error, while larger values of λ improve stability but cause underfitting. This is consistent with the bias-variance trade-off: regularization reduces sensitivity and variance, but excessive regularization increases bias.

The dataset-size experiment provides the strongest empirical support for the stability theory. Increasing the training sample size from 50 to 320 reduces prediction and loss. The value of n multiplied by prediction change stays consistent across the experiment, which is consistent with an approximate $1/n$ relationship.

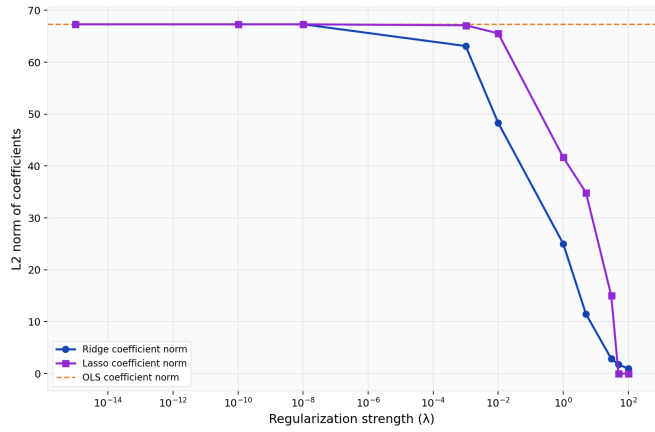


(a) Stability vs test error for the Diabetes dataset.

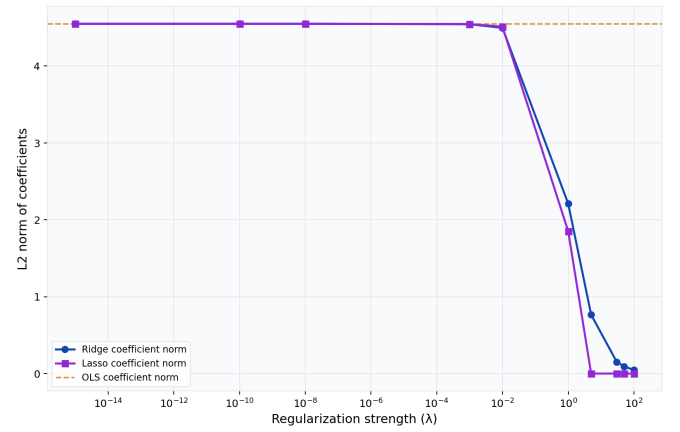


(b) Stability vs test error for the synthetic regression dataset.

Fig. 4: Relationship between empirical stability and test error for Ridge and Lasso.

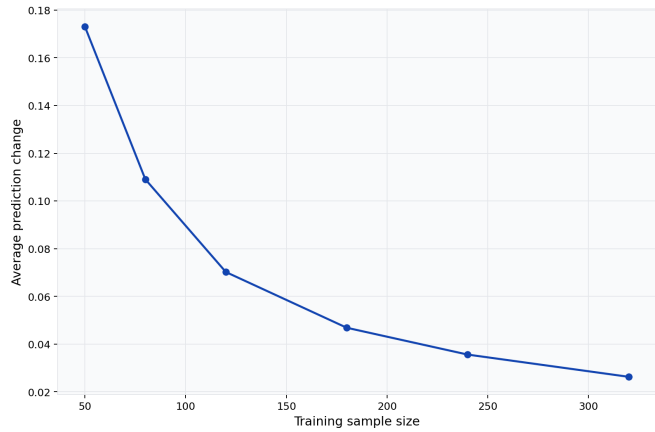


(a) Model complexity for the Diabetes dataset.

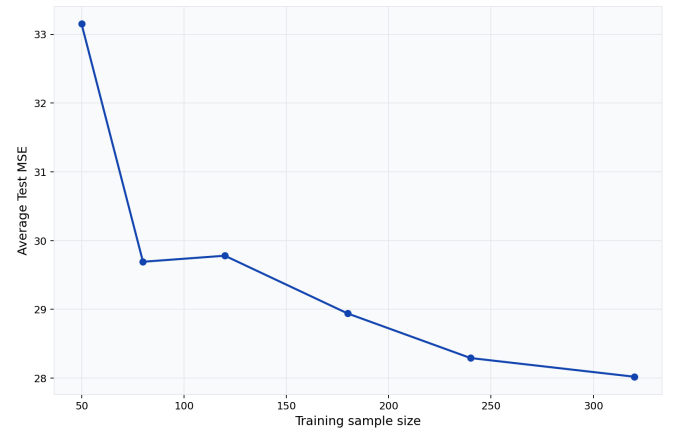


(b) Model complexity for the synthetic regression dataset.

Fig. 5: Effect of regularization on model complexity, measured by the coefficient norm.



(a) Effect of dataset size on stability.



(b) Effect of dataset size on test error.

Fig. 6: Effect of training sample size on stability and test error for the synthetic dataset.

Overall, the results support the stability-based interpretation of regularization, but they also show its practical limitation.

The goal is not to choose the largest possible λ , nor simply the most stable model. The best model is the one that balances

predictive accuracy, stability, and model complexity.

VII. CONCLUSION

This report studied the relationship between regularization, empirical stability, and generalization in linear regression. The results show that Ridge regression can reduce sensitivity to individual training observations as λ increases from very small to moderate values, supporting the stability-based interpretation of regularization.

However, the experiments also show that the most stable model is not always the most accurate one. Very large values of λ shrink the coefficients too strongly, causing underfitting and higher test error. Therefore, the best model is not obtained by maximizing regularization, but by balancing stability and predictive accuracy.

The additional Lasso comparison shows that L_1 regularization can reduce model complexity even more aggressively by setting some coefficients to zero, but the main stability interpretation in this report remains based on Ridge regression. Overall, Ridge regression provides a clear example of how regularization connects the classical bias–variance trade-off with the algorithmic stability view of generalization.

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